

	GLOBAL ECONOMICS COMPUTATIONAL THINKING GRADE: 10TH TERM 2 – PROJECT JAVASCRIPT DECISION ANALYSIS PROJECT Nicolás López Cuéllar, Jaime David Navarrete	SECOND TERM 2025–2026
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Learning objective: Present and defend a JavaScript tool that evaluates all possibilities in a decision-making problem using the strategies presented.	Assessment criteria: Global Economics C10: Evaluates all possibilities in a decision-making problem using the strategies presented. Computational Thinking C8: Designs interactive elements in web page for a purpose. C9: Creates functions in a web page for a purpose. C10: Evaluates interactivity in web pages based on requirements.
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1. Project brief

1.1 Purpose of the project

This programming project requires students to design and implement a JavaScript tool that automates C8 decision-analysis calculations (Bayes/Expected Value, Maximin, and Minimax Regret). The results produced by this tool must then be used to support a C9 decision-tree conclusion. The presentation and defense of the implemented program constitute the evidence for Criterion C10.

1.2 Input specification

The program must accept a single payoff table containing:

- States of nature with their corresponding probabilities.
- A set of decision alternatives.
- The payoff for each alternative–state pair.

Example input format (values in thousand USD):

State	Probability	A_1	A_2	A_3	A_4	A_5
S_1	0.40	140	130	155	148	138
S_2	0.35	95	105	90	110	100
S_3	0.25	60	80	45	85	72

1.3 Required Outputs

For each alternative, the program must compute:

- Maximum payoff (for Maximax).
- Minimum payoff (for Maximin).
- Expected value using the given probabilities (Bayes criterion).

- Maximum regret (Minimax Regret criterion).

The results must be displayed in a structured summary table such as:

Alt.	Max P.	Min P.	EV	Max Regret
A_1	$\max P(A_1)$	$\min P(A_1)$	$EV(A_1)$	$\max R(A_1)$
A_2	$\max P(A_2)$	$\min P(A_2)$	$EV(A_2)$	$\max R(A_2)$
A_3	$\max P(A_3)$	$\min P(A_3)$	$EV(A_3)$	$\max R(A_3)$
A_4	$\max P(A_4)$	$\min P(A_4)$	$EV(A_4)$	$\max R(A_4)$
A_5	$\max P(A_5)$	$\min P(A_5)$	$EV(A_5)$	$\max R(A_5)$

1.4 Decision workflow (C10 → C8 → C9)

1. **(C10)** Execute the JavaScript program to generate the structured summary table.
2. **(C8)** Identify and justify recommendations using the computed results.
3. **(C9)** Use the selected C8 recommendations as the basis for constructing the C9 decision tree.

After presenting the program and its outputs, students must apply this workflow to the assigned project problems in order to activate Criteria C8 and C9.

2. Solved Example 1

1. Program-Based Ranking and Structured Decision Refinement

A firm must choose one strategy among A_1, A_2, A_3, A_4, A_5 under uncertain market conditions.

State	Probability	A_1	A_2	A_3	A_4	A_5
H	0.45	150	142	165	155	148
M	0.35	98	112	90	118	104
L	0.20	52	78	35	70	88

After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Step 1: Retain the top three alternatives according to Expected Value (EV).
- Step 2: Within that subset, eliminate the alternative with the highest Max regret.
- Step 3: From the remaining alternatives, select the one with the highest minimum payoff (Maximin).

C10 summary table (program output)

Alt.	Max P.	Min P.	EV	Max regret
A_1	150	52	112.20	36
A_2	142	78	118.70	23
A_3	165	35	112.75	53
A_4	155	70	125.05	18
A_5	148	88	120.60	17

C8 solution (ranking and recommendations)

- EV ranking:
 $A_4 > A_5 > A_2 > A_3 > A_1$.
- Maximax ranking:
 $A_3 > A_4 > A_1 > A_5 > A_2$.
- Maximin ranking:
 $A_5 > A_2 > A_4 > A_1 > A_3$.
- Minimax Regret ranking:
 $A_5 < A_4 < A_2 < A_1 < A_3$.

C9 solution (apply filters)

- (Step 1) EV top 3: $\{A_4, A_5, A_2\}$.
- (Step 2) Lowest two regrets: $\{A_4, A_5\}$.
- (Step 3) Maximin selects A_5 .
- Final decision: A_5 .

3. Solved Example 2

Multi-Criteria Filtering and Sequential Decision Selection

An insurance broker considers five portfolio mixes Y_1 to Y_5 under three climate-risk states. Payoffs are in thousand USD.

State	Prob.	Y_1	Y_2	Y_3	Y_4	Y_5
W_1	0.40	136	126	146	116	130
W_2	0.35	94	98	84	103	90
W_3	0.25	66	74	58	88	69

After obtaining the C10 summary and computing the C8 results, apply the following structured C9 decision procedure:

- Step 1: Retain the top three alternatives according to Expected Value (EV).
- Step 2: Within that subset, apply the Minimax Regret criterion and keep the top two performing options.
- Step 3: Then, apply the Maximin criterion to determine the final choice.

C10 summary table (program output)

Alt.	Max P.	Min P.	EV	Max regret
Y_1	136	66	103.80	22
Y_2	126	74	103.20	20
Y_3	146	58	102.30	30
Y_4	116	88	104.45	30
Y_5	130	69	100.75	19

C8 solution (ranking and recommendations)

- EV ranking:
 $Y_4 > Y_1 > Y_2 > Y_3 > Y_5$.
- Maximax ranking:
 $Y_3 > Y_1 > Y_5 > Y_2 > Y_4$.
- Maximin ranking:
 $Y_4 > Y_2 > Y_5 > Y_1 > Y_3$.
- Minimax Regret ranking:
 $Y_5 < Y_2 < Y_1 < Y_3 = Y_4$.

C9 solution (apply filters)

- (Step 1) EV top 3: $\{Y_4, Y_1, Y_2\}$.
- (Step 2) Minimax Regret keeps $\{Y_2, Y_1\}$.
- (Step 3) Maximin selects Y_2 .
- Final decision: Y_2 .